

Calibration Report

Human - IH_powerLaw_stressBased

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Excellent

QUALITY

1.00

WORST R-HAT

4400

MIN ESS

Run Configuration

Species	human
Model	IH_powerLaw_stressBased
Control variables	shear_stress, exposure_time
Likelihood	normal
MCMC walkers	100
Burn-in steps	2500
Production steps	2500
Calibrate noise	Yes

CALIBRATED PARAMETERS

PARAMETER	DISTRIBUTION	BOUND / LOCATION	BOUND / SCALE
A	Uniform	1e-07	20.0
alpha	Uniform	0.5	3.0
beta	Uniform	0.01	1.0
sigma	Uniform	1e-09	0.5

MCMC Diagnostics

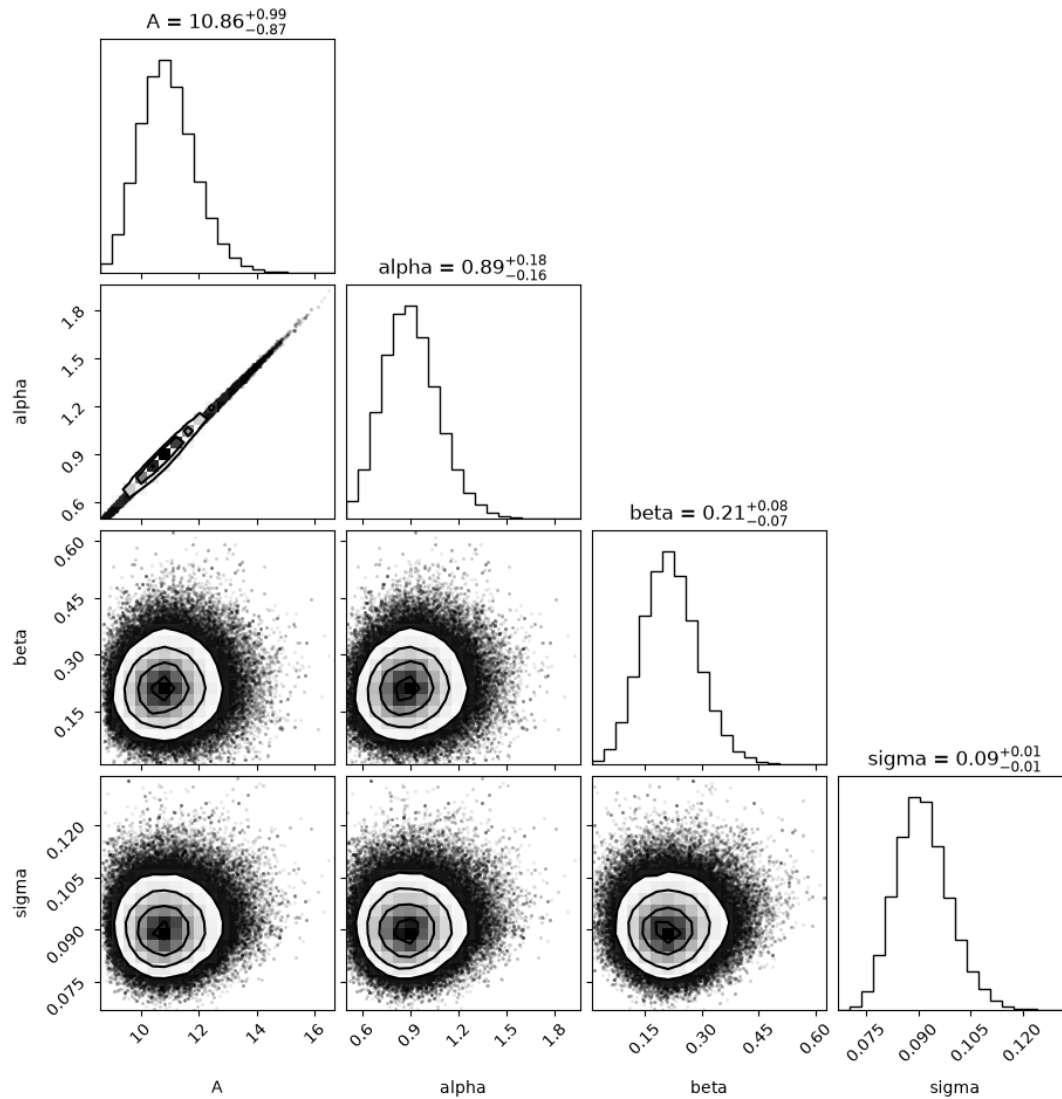


Figure 1 - Pairwise posterior corner plot.

Table 1 - Convergence diagnostics (post burn-in, 100 walkers).

PARAMETER	MEAN	SD	ESS BULK	ESS TAIL	R-HAT	ESS QUALITY	R-HAT QUALITY
A	11.0000	0.9500	5300	10000	1.00	Excellent	Excellent
alpha	0.9000	0.1700	5300	10000	1.00	Excellent	Excellent
beta	0.2200	0.0730	4400	9600	1.00	Excellent	Excellent
sigma	0.0910	0.0078	5400	10000	1.00	Excellent	Excellent

R-hat thresholds: < 1.01 = Excellent, 1.01-1.10 = Good, >= 1.10 = Poor.

ESS thresholds: > 400 = Excellent, 101-400 = Acceptable, <= 100 = Poor.

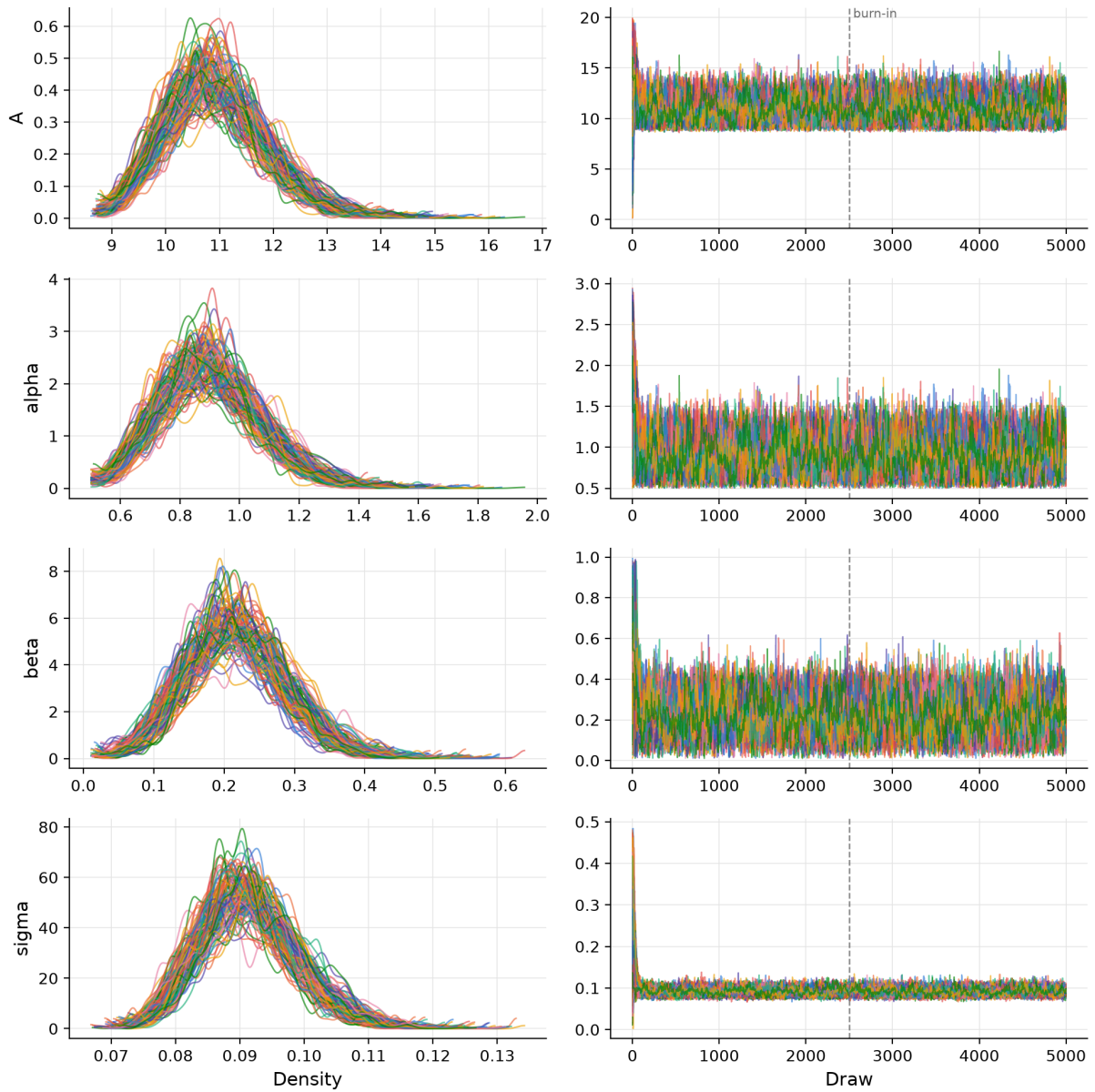


Figure 2 - Trace plots. Left: post burn-in posteriors; right: sample traces per walker (burn-in marked).

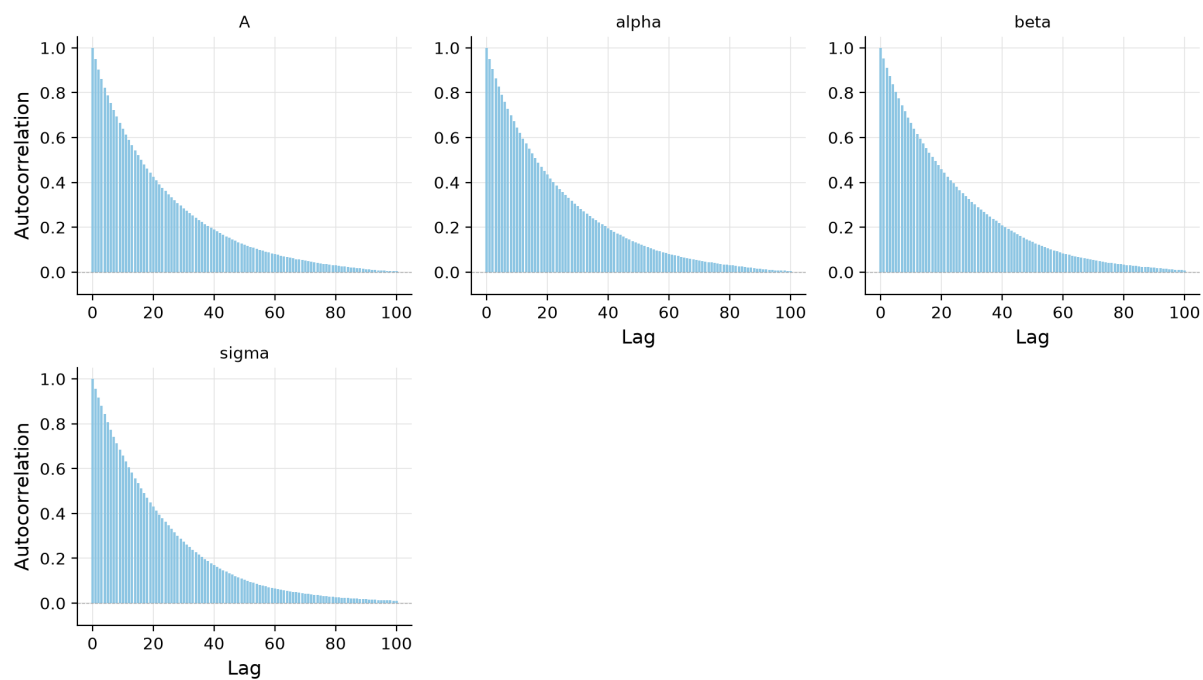


Figure 3 - Autocorrelation by lag for each parameter.